

TD3 vs DDPG Performance Analysis in Autonomous Driving Environment

Structured research-style report built from seed-aggregated logs, grouped noise-level plots, algorithm comparisons, and summary text.

Project Title	TD3 vs DDPG Performance Analysis in Autonomous Driving Environment
Description	Comparison of reward systems across noise levels and algorithm behavior in a driving environment.
Generated	2026-06-30 15:58:35

Summary Metrics

Metrics are reported as mean $\pm$ standard deviation where available. Crash rate is treated as a primary safety metric, and lower convergence episode indicates faster sample-efficient learning.

Experiment	TD3 Reward $\pm$ Std	DDPG Reward $\pm$ Std	TD3 Crash $\pm$ Std	DDPG Crash $\pm$ Std	TD3 Conv	DDPG Conv
R1_N1	0.57 $\pm$ 2.24	0.36 $\pm$ 2.70	89.33 $\pm$ 11.73%	91.33 $\pm$ 6.18%	32 $\pm$ 15	36 $\pm$ 20
R1_N2	1.28 $\pm$ 0.43	1.54 $\pm$ 3.25	96.00 $\pm$ 2.94%	84.67 $\pm$ 18.26%	65 $\pm$ 7	70 $\pm$ 7
R1_N3	-2.66 $\pm$ 0.45	-0.78 $\pm$ 1.80	100.00 $\pm$ 0.00%	89.67 $\pm$ 14.61%	118 $\pm$ 41	115 $\pm$ 31
R2_N1	16.68 $\pm$ 6.50	18.80 $\pm$ 16.44	76.33 $\pm$ 33.47%	76.00 $\pm$ 26.55%	34 $\pm$ 15	80 $\pm$ 18
R2_N2	19.01 $\pm$ 7.76	6.01 $\pm$ 0.86	97.00 $\pm$ 2.94%	100.00 $\pm$ 0.00%	55 $\pm$ 20	86 $\pm$ 24
R2_N3	6.45 $\pm$ 1.36	6.19 $\pm$ 3.65	100.00 $\pm$ 0.00%	100.00 $\pm$ 0.00%	106 $\pm$ 58	103 $\pm$ 38
R3_N1	14.89 $\pm$ 10.07	30.01 $\pm$ 4.59	100.00 $\pm$ 0.00%	79.00 $\pm$ 25.46%	91 $\pm$ 50	35 $\pm$ 18
R3_N2	8.52 $\pm$ 13.73	21.47 $\pm$ 10.42	30.67 $\pm$ 43.37%	99.33 $\pm$ 0.47%	101 $\pm$ 34	54 $\pm$ 22
R3_N3	13.37 $\pm$ 2.94	17.18 $\pm$ 4.93	99.67 $\pm$ 0.47%	99.33 $\pm$ 0.94%	108 $\pm$ 49	77 $\pm$ 25
R4_N1	17.56 $\pm$ 10.98	25.02 $\pm$ 11.77	66.33 $\pm$ 46.91%	95.67 $\pm$ 4.19%	33 $\pm$ 16	55 $\pm$ 25
R4_N2	24.73 $\pm$ 21.26	27.79 $\pm$ 5.14	93.33 $\pm$ 9.43%	99.33 $\pm$ 0.94%	100 $\pm$ 43	60 $\pm$ 29
R4_N3	38.32 $\pm$ 9.45	51.31 $\pm$ 20.58	92.33 $\pm$ 8.81%	78.67 $\pm$ 29.47%	59 $\pm$ 4	78 $\pm$ 19

Analysis by Noise Level

Each subsection fixes one noise level and compares R1-R4 for TD3 and DDPG using the grouped plots generated from seed-aggregated curves.

Noise Level N1

Grouped experiment curves compare R1-R4 within a fixed noise setting. Each plot shows the seed-aggregated mean with light uncertainty shading.



Summary

Noise N1: higher reward tends to appear in R3, while stronger stability tends to appear in R1. TD3 reward 12.43 ± 6.91 , DDPG reward 18.55 ± 11.22 ; TD3 crash $83.00 \pm 12.76\%$, DDPG crash $85.50 \pm 8.21\%$.

Key Takeaways

- Safety must be interpreted jointly with reward; high reward and high crash is unsafe.
- Lower standard deviation indicates more stable learning under this noise level.

Noise Level N2

Grouped experiment curves compare R1-R4 within a fixed noise setting. Each plot shows the seed-aggregated mean with light uncertainty shading.



Summary

Noise N2: higher reward tends to appear in R4, while stronger stability tends to appear in R1. TD3 reward 13.38 ± 9.09 , DDPG reward 14.20 ± 10.78 ; TD3 crash $79.25 \pm 28.08\%$, DDPG crash $95.83 \pm 6.45\%$.

Key Takeaways

- Safety must be interpreted jointly with reward; high reward and high crash is unsafe.
- Lower standard deviation indicates more stable learning under this noise level.

Noise Level N3

Grouped experiment curves compare R1-R4 within a fixed noise setting. Each plot shows the seed-aggregated mean with light uncertainty shading.



Summary

Noise N3: higher reward tends to appear in R4, while stronger stability tends to appear in R1. TD3 reward 13.87 ± 15.22 , DDPG reward 18.48 ± 20.01 ; TD3 crash $98.00 \pm 3.27\%$, DDPG crash $91.92 \pm 8.67\%$.

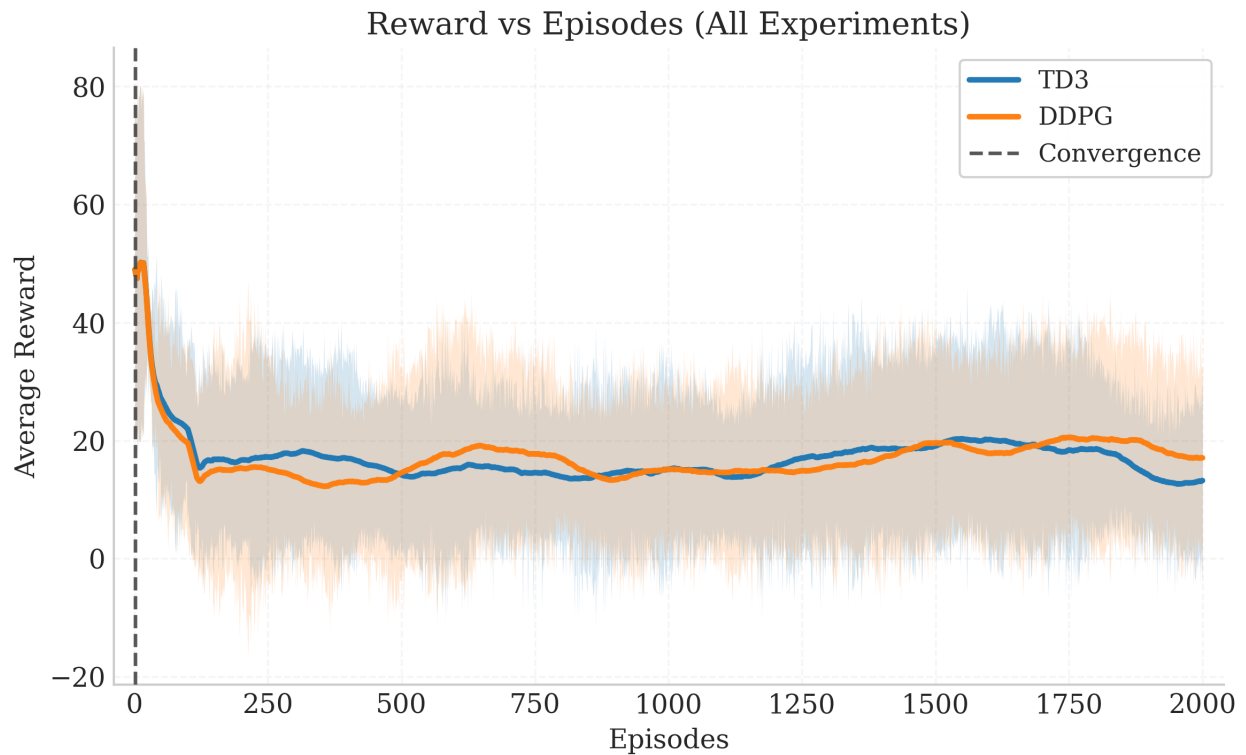
Key Takeaways

- Safety must be interpreted jointly with reward; high reward and high crash is unsafe.
- Lower standard deviation indicates more stable learning under this noise level.

Algorithm Comparison

The next pages compare TD3 and DDPG directly using the existing comparison plots, focusing on convergence speed, stability, and final performance.

TD3 vs DDPG Reward Comparison



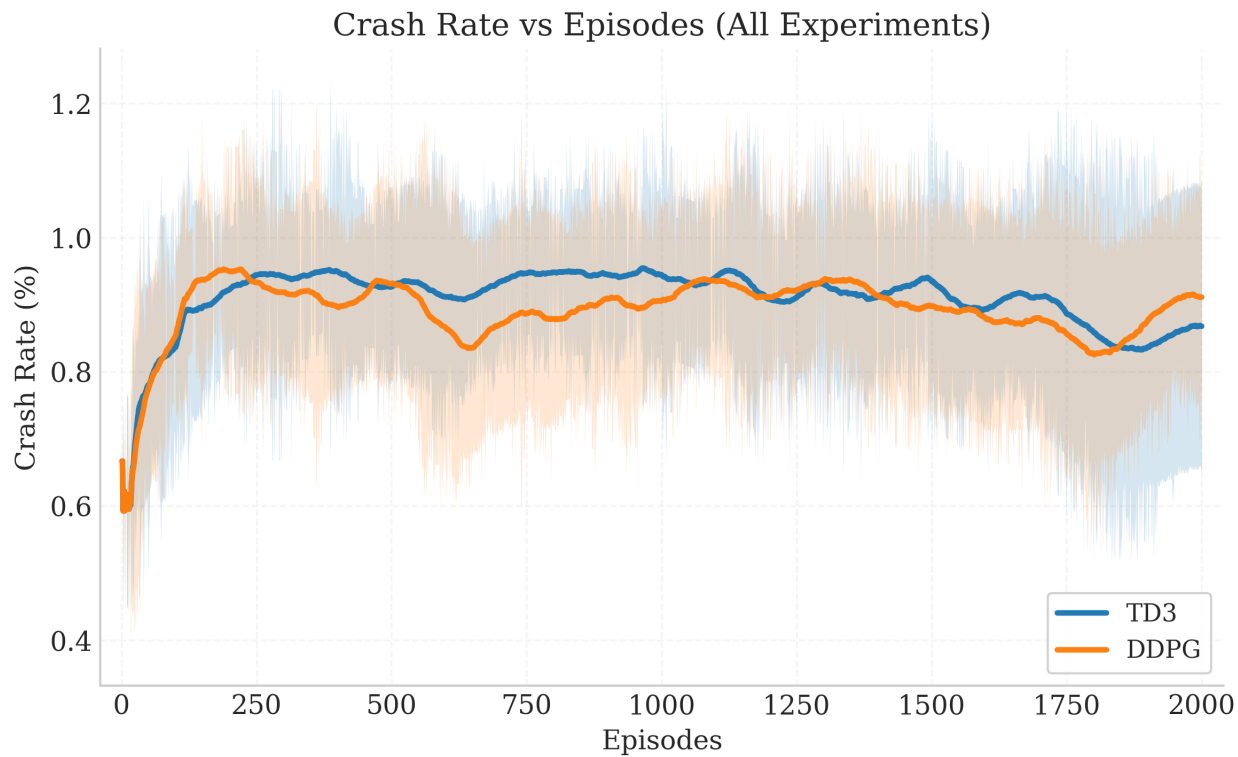
Caption. Reward vs Episodes across all experiments with uncertainty represented as mean $\pm$ standard deviation.

TD3 reward 13.23 ± 11.00 , DDPG reward 17.07 ± 14.78 ; TD3 crash $86.75 \pm 19.66\%$, DDPG crash $91.08 \pm 8.92\%$; TD3 convergence 75 ± 31 , DDPG convergence 71 ± 23 .

Key Takeaways

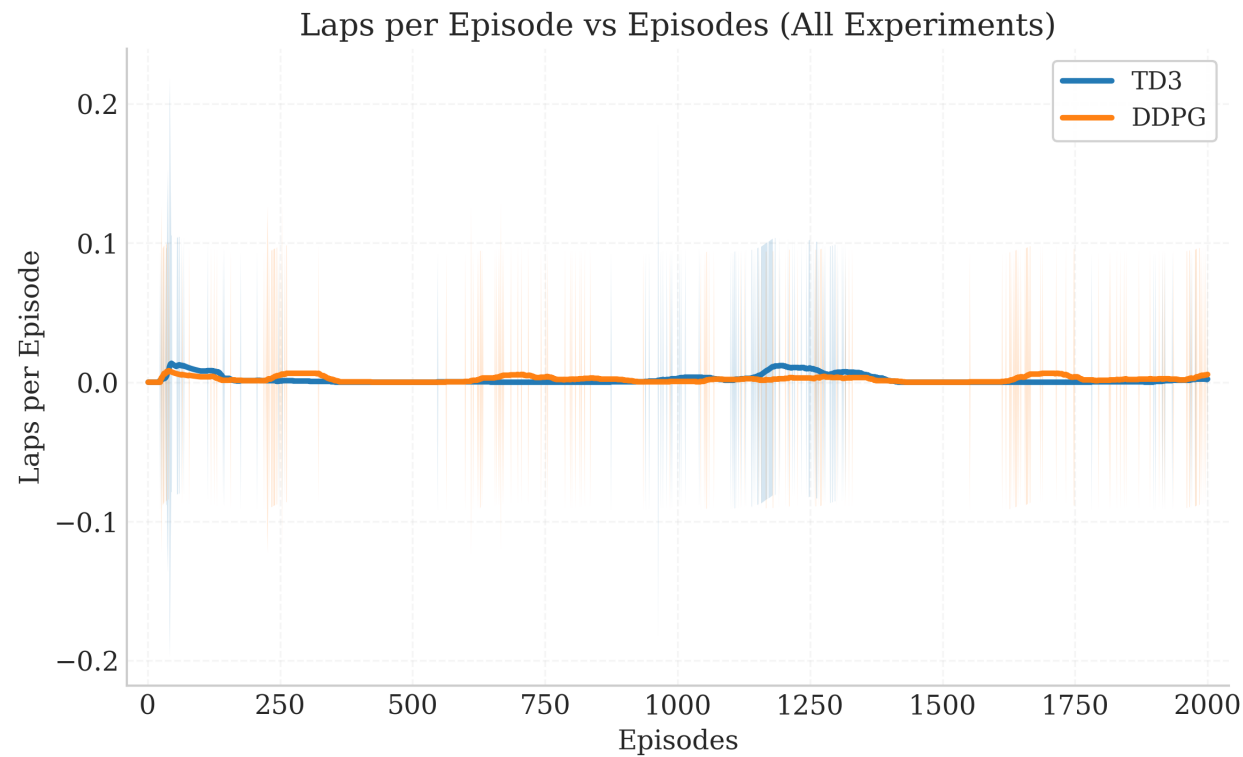
- Lower convergence episode implies faster sample-efficient learning.
- Reward must be interpreted with crash rate to avoid unsafe conclusions.

TD3 vs DDPG Crash Rate Comparison



Caption. Crash Rate vs Episodes across all experiments with uncertainty represented as mean $\pm$ standard deviation.

TD3 vs DDPG Laps Comparison



Caption. Laps vs Episodes across all experiments with uncertainty represented as mean $\pm$ standard deviation.

Key Aggregate Insights

The following section summarizes cross-noise trade-offs with explicit safety, stability, and sample-efficiency interpretation.

Highest reward-system trend across noise levels: R4. Most stable reward-system trend: R1. Lower crash tendency appears in TD3, while faster convergence appears in DDPG. Noise generally reduces reward and increases variability, emphasizing a trade-off between raw performance and robustness.

Key Takeaways

- Stability-sensitive deployments should prioritize low variance and low crash behavior.
- Performance-oriented deployments should still verify safety under higher noise.

Final Conclusion

The results indicate a trade-off between performance and robustness. Highest reward trend is associated with R4, while greater stability trend is associated with R1. DDPG can achieve higher rewards and faster convergence in several settings, while TD3 can provide more stable learning and improved safety in certain conditions. Higher noise levels generally reduce reward and increase instability.

Key Takeaways

- TD3 is preferable when stability and safety are prioritized.
- DDPG is preferable when higher reward is prioritized and crash behavior remains acceptable.